Bike Sharing Predictive Analysis

To: Hypothetical Bike-Sharing Operations Manager
From: Anshuman
Subject: Historical forecasting evidence and a proposed pilot

Question

Can next-day hourly forecasts improve aggregate rental-service capacity planning over repeating last week? At 11:59 PM, the proposed workflow predicts each of the next calendar day's 24 hourly rental counts. The decision is a system-wide hourly throughput plan to inform staffing review. This dataset cannot determine how many bicycles to place at a particular station.

Data and Model

The public UCI Bike Sharing dataset contains 17,379 observed hourly counts from Capital Bikeshare in 2011-2012. Its 165 missing hourly records remain missing; no rental observations were invented. Inputs are calendar information and demand lags known before issuance. We exclude casual and registered rentals because they compose the target, and exclude future observed weather because no weather-forecast archive exists. A Ridge regression pipeline and a tuned gradient-boosting pipeline are compared with the same-hour-last-week baseline. Training-only preprocessing and time-ordered cross-validation isolate all fitting from testing.

Honest Evaluation

Model fitting ends October 7, 2012; October 8-November 4 is reserved for interval calibration. The final eight complete weeks, November 5-December 30, contain 1,302 comparable scored hours. Four missing actuals and 38 missing baseline references are excluded. Baseline MAE is 71.95 rentals per hour, versus 64.10 for Ridge and 69.30 for tuned boosting. RMSE is 119.50, 96.10, and 108.42, respectively. Ridge has the lowest error; the more complex model does not win. These are daily rolling-origin feature updates with frozen fitted models, not one forecast made eight weeks in advance. Missing-history exclusions and the autumn/winter test window limit generalization.

Decision and Money

The original point-capacity scenario assumes $5 per underplanned rental-unit and $1 per excess unit: baseline $230,005, boosting $133,389 over 1,302 hours. The new Gurobi model instead chooses integer assignments across six four-hour shifts, capped at 20 assignments per shift and 48 per day. It assumes 40 rental-service units per worker-hour and $20 hourly pay, minimizing wages plus forecast shortage/excess penalties. These are hypothetical throughput assignments, not identified employees or station allocations.

All 144 model-based solves reached certified optimality. On 48 complete common days (1,152 hours), actual-evaluated proxy costs are seasonal $401,486, Ridge $372,872, and boosting $380,462; flat staffing costs $546,974. Actual targets never enter the solver. Eight partial days are excluded equally. Sensitivity revalues the same plans under nine cost combinations. This extension was designed after forecast scores were known: it is exploratory retrospective analysis, not a fresh holdout or causal savings. Completed rentals omit unmet demand; validate throughput, wages and constraints prospectively.

Trust Story

For the high-demand case at 2012-11-07 17:00:00, the model forecasts 835.2 rentals; the largest local contributions are lag_168 (+280.1 rentals) and trend_days (+96.1 rentals). These inputs move this prediction relative to the model reference of 187.8 rentals; they describe the model's reasoning, not the causal effect of changing capacity or the calendar.

Three genuine SHAP waterfalls explain low, moderate, and high forecasts. Chronologically calibrated MAPIE intervals target 90% coverage but achieve only 85.64%, with average width 251.4 rentals. Serial dependence and changing conditions mean ordinary exchangeability-based guarantees do not apply automatically. The worst boosting slice is hour=17, with MAE 174.7 and RMSE 210.1 across 55 hours. Weather changes, special events, missing history, reporting delays, and demand shifts remain important failure modes.

Before deployment, test the frozen decision rule prospectively and confirm that counts are available by issuance. Monitor missing hours, weekly MAE and RMSE against the concurrent baseline, commute-hour errors, interval coverage and width, and assumed-cost outcomes. Review sustained coverage below 90% or errors above baseline for two weeks; recalibrate or retrain using chronological data. Obtain station inventory and unmet-demand evidence before making station allocation claims. This project is a research prototype; it has not been deployed or submitted. OpenAI Codex assisted with coding, analysis and drafting; Anshuman is responsible for reviewing and explaining the work.

Source: Fanaee-T & Gama (2013), Event labeling combining ensemble detectors and background knowledge, doi:10.1007/s13748-013-0040-3. UCI Bike Sharing dataset, dataset 275. Full methods, plots, assumptions and checks accompany the executed notebooks.